CRT Chapter 5

1.A loop repeats a code until it met what it wanted

2. A ‘while’ loop checks before running. A ‘do-while’ loop runs the code at least once before before checking the condition

3. The review section on input validation loops.

4.

a) A loop that never stops
b) The loop condition never becomes false, and the counter is not updated correctly.
c) Overflow occurs when a variable passes the largest value it can store.

5. 60 times, because the value of ‘x’ increases by 2 each time until it reaches 120.

6.any value of ‘x’ that is 120 or higher. The loop adds 3, so the condition will stay false after the first iteration.

7.Counter: increases by a set amount. Uses: counting loop on repeat and counting students.
Accumulator: Adds values to a running total. Uses: adding grades and calculating total sales.

8.

```
Int sum = 0;

For (int i = 3; i <= 10; 1++) {
        Sum += i;
}
System.out.print(sum);
```

9. Whether the loop must execute at least once.
Whether the number of repetitions is known.

11.

a) 9
b) “My “(the first three characters, including the space).
c) “My string”
d) “MY STRING”

e) “My string”(there are no space at the beginning or end).